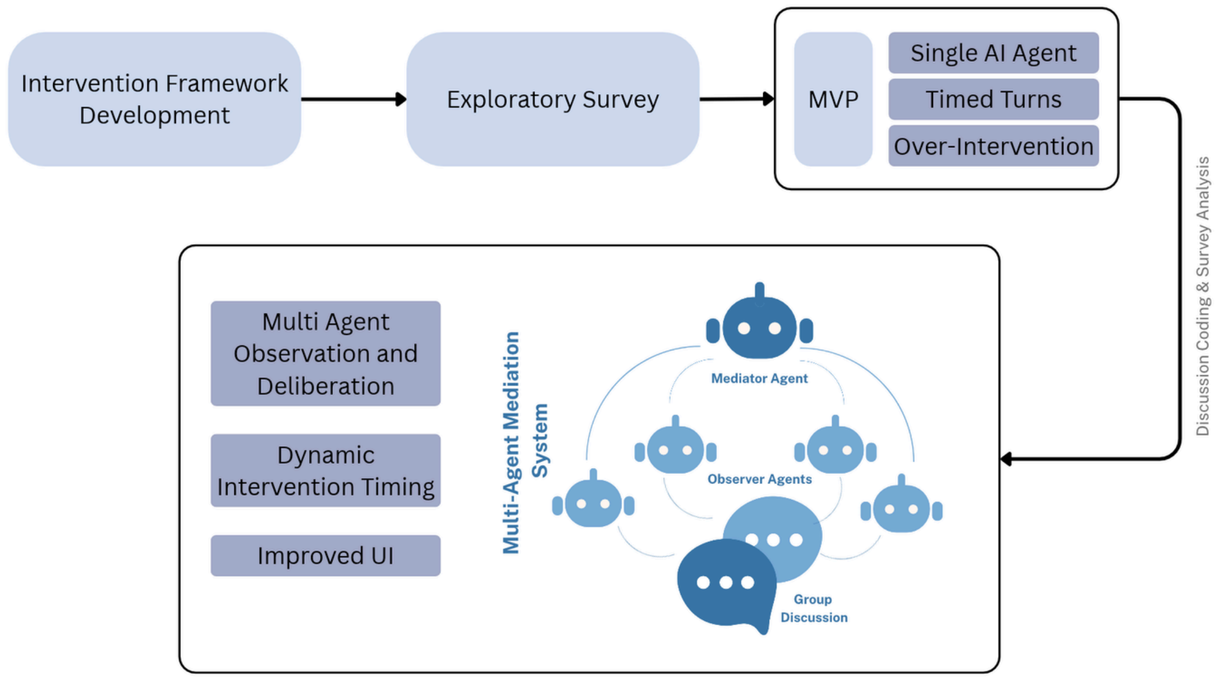
AI based Mediation for Collaborative Decision Making under Hidden Profile Conditions

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Introduction

Group decision making is widely used on the assumption that combining members' knowledge leads to better outcomes, but it is often limited by biases and communication dynamics that shape which members contribute and which information is discussed [1]. Professional mediators can address these failures, but they are costly and scarce, limiting their use in everyday group settings. With increasing nubers of the wider public now using AI tools [2], large language models (LLMs) have become increasingly present in team and collaborative contexts, raising the possibility of their use in a mediating role. This project investigates the implementation of an LLM-based mediator in small group discussions under hidden profile conditions [3], examining its effect on group decision dynamics alongside participants' trust, perception and acceptance of the tool.

Iterative Development



The Hidden Profile Paradigm

First established by Stasser and Titus in 1985 [2], the Hidden Profile problem illustrates the effects of bias on group discussions. When information is distributed asymmetrically across a group, members tend to gravitate towards discussing shared information, discounting new facts that may go against their original preferences. In their original paper, nearly 70% of groups failed to reach the correct solution under these conditions and nearly 40 years of research confirms this finding [1].

Intervention 1:

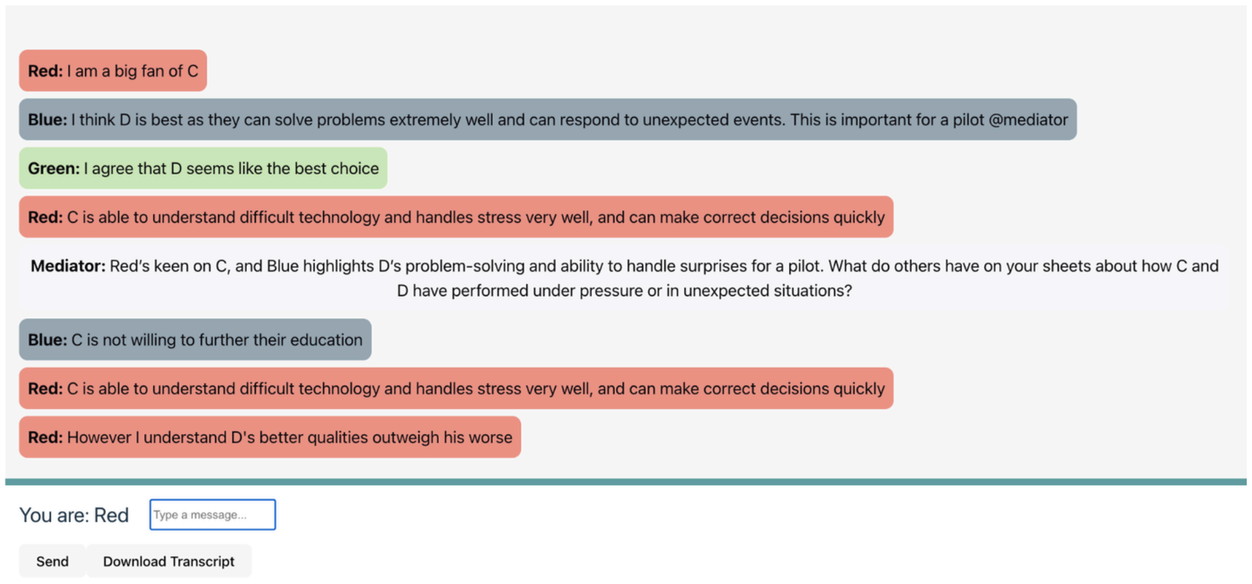
Convergence Agent: 0.2
Equal Consideration Agent: **0.15**
Equal Participation Agent: 0.5
Group Information Agent: 0.6

→ **Equal Consideration Agent:** "The conversation is narrowly focused on Stevens: Green repeatedly raises negative points (discourages new ideas, lack of fundraising), Red offers a brief defense, and Blue speculates coworkers are undermining him; alternative candidates or solutions are not discussed."

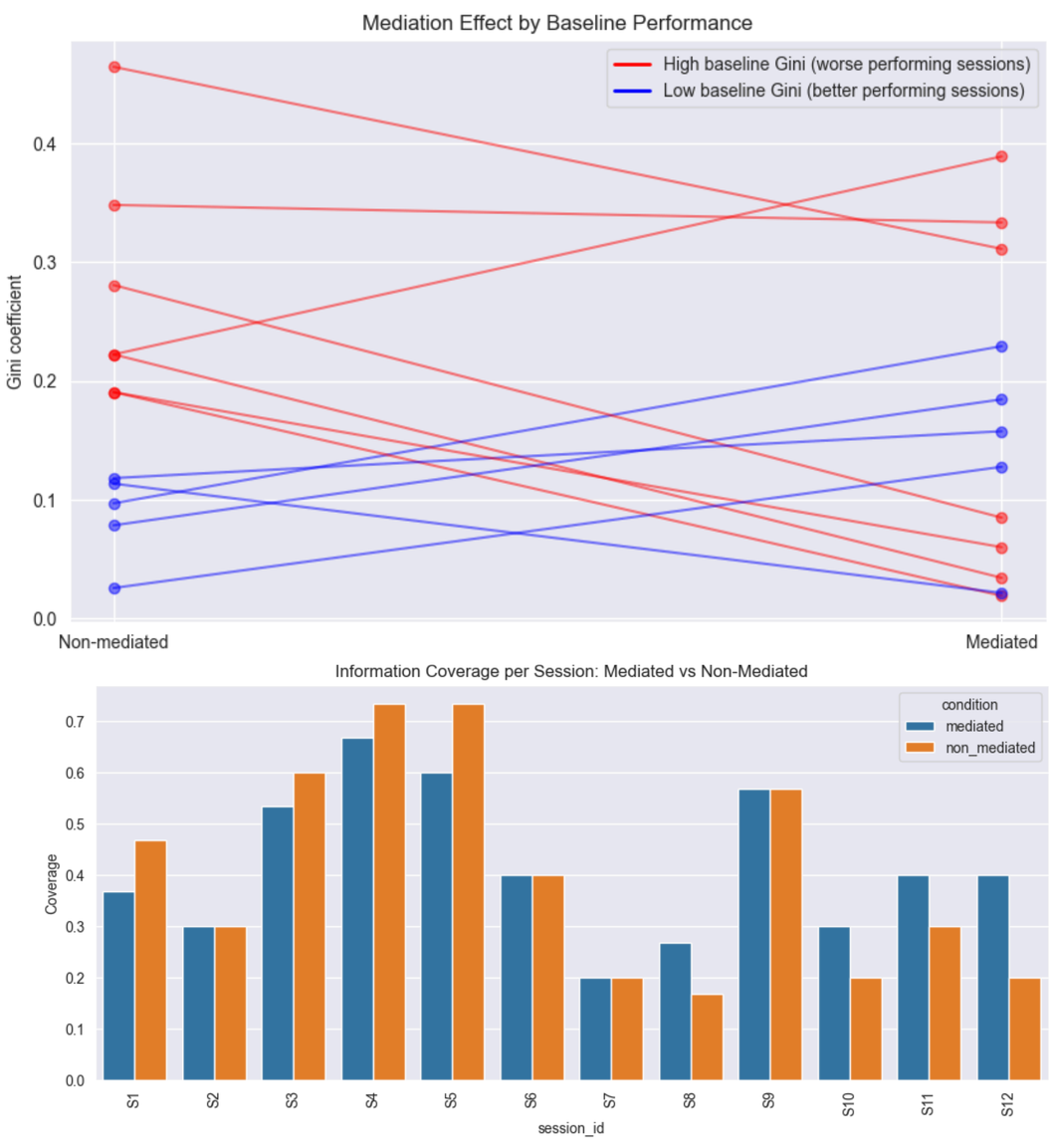
→ **Mediator Message:** "We've mostly been weighing Stevens's issues with new ideas and fundraising, and Blue also mentioned Jones. Does anyone have anything on Jones's track record with fundraising or how he handles new ideas?"

Study Design

Each group of participants took part in 2 hidden profile discussion scenarios across two conditions: one in which the AI mediator was present and active and one in which it was not. This allowed for direct comparison of mediated and unmediated conversation within the same participant groups, controlling for individual and group-level variation in communication style. To mitigate learning effects, the assignment of conditions to scenarios was counterbalanced across groups. Users were able to connect remotely to the site and complete the discussion via a text based interface.



Results



Mediator Perception

- **Mediation as Information**
- **Leader vs Assistant Perceptions**
- **External to Group Dynamic**

Effects of Previous AI Use

- **Recognisably AI Style**
- **Hallucinations & Domain Trust**
- **Limitations of Relationship Building**

Interaction Flow

- **Behind the Timing of the Group Conversation**
- **Lack of Wider Context**

Conclusions

- Improved Collaboration: The mediator improved information coverage and participation equality in lower-performing groups, while providing useful structure and prompting information sharing.
- Technical Limitations: Processing delays and limited contextual awareness sometimes disrupted conversational flow, while the mediator struggled to adapt to changing discussion stages.
- Trust & Acceptance: Although its neutral and balanced approach was valued, concerns around AI reliability, communication style and its perceived role as an external group member highlighted challenges to wider acceptance.

Future work could explore alternative LLMs for speed and improved contextual awareness to reduce delays and improve conversational flow. Further development could investigate adapting interventions to different conversational stages based on the temporal nature of conversation development and refining observer-agent mechanisms. Larger-scale studies in natural group settings could also examine user trust, acceptance and the effectiveness of AI mediation across different group dynamics.

[1] *Li Lu, Y. Connie Yuan, and Poppy McLeod. Twenty-five years of hidden profiles in group decision making: A meta-analysis. Personality and social psychology review : an official journal of the Society for Personality and Social Psychology, Inc, 16:54–75, 09 2011. doi:10.1177/1088868311417243.*

[2] Nicole Gillespie, Steven Steven Lockey, Tabi Ward, Alexandria Macdade, and Gerard Hassed. Trust, attitudes and use of artificial intelligence: A global study 2025. Technical report, April 2025.

[3] Garold Stasser and William Titus. Pooling of unshared information in group decision making: Biased information sampling during discussion. Journal of Personality and Social Psychology, 48(6):1467–1478, Jun 1985. doi:10.1037/0022-3514.48.6.1467.